BUDGET WATCHDOG PLATFORM

AI-Powered Government Spending Intelligence

₹2.1L Cr

Budget Lapse
Annually (India)

68%

Debts Miss
Spend Targets

>0 %

Real-Time
Watchdog Systems

THE PROBLEM

₹2.1 Lakh Crore disappears every year — and nobody catches it in time.

Budget Lapse

₹2.1L Cr unspent annually. CAG reports show 30–40% of scheme budgets lapse in final quarter — money meant for roads, schools, hospitals never reaches beneficiaries.

40%
avg lapse rate

Corrupt Patterns

Year-end dumping, micro-transaction splitting, ghost departments. These patterns exist in data but no automated system flags them until audit — 18 months later.

18 mo
avg detection lag

No Prediction

Officials have no early warning. When fiscal year end arrives with 60% budget unspent, it's too late to reallocate. ₹847 Cr in Maharashtra alone lapsed in FY2023.

₹847Cr
Maharashtra FY23 lapse

OUR SOLUTION

One platform. Four jobs. Zero budget leakage.

01

Track Money Flow

Live scoreboard: allocated vs spent vs remaining per department per district. 30-second auto-refresh.

02

Detect Anomalies

Isolation Forest ML + 6 hard rules flag burst spending, ghost debts, year-end dumping, micro-splits instantly.

03

Predict Lapse Risk

Weighted linear extrapolation warns months in advance which departments will leave budget unspent by FY end.

04

Suggest Reallocation

Matching algorithm moves predicted-unspent funds from low-spend depts to underfunded ones — fully auditable.

TECH STACK & ARCHITECTURE

Frontend	React + TypeScript	5 pages · React Query (30s refresh) · Recharts · Zustand · Tailwind
Backend / API	FastAPI (Python)	16 REST endpoints · Real-time transaction response · Pydantic validation
Database	PostgreSQL	6 tables · Append-only transactions · All derived fields computed at runtime
ML / Analytics	scikit-learn — Isolation Forest	9 features · Unsupervised · No labels needed · Weekly retrain
Background Jobs	Celery + Redis	6-hour full pipeline · Sunday 2am model retrain · Event-driven triggers
Infra	Docker Compose	Single command deploy · All services containerized · Production-ready

HOW IT WORKS — END TO END

1
Officer Logs Transaction

2
API Triggers Anomaly Check

3
ML + Rules Score Dept

4
Alert Returns in Same Response

AUTOMATED PIPELINE SCHEDULE

Trigger	Stage 1: Anomaly	Stage 2: Lapse	Stage 3: Realloc
User adds transaction	✅ That debt only (instant)	✅ That debt only	❌
Every 6 hours (Celery)	✅ All departments	✅ All departments	✅ All
Sunday 2am	✅ + Model retrain	✅	✅
Manual API trigger	✅	✅	✅

FEASIBILITY & VIABILITY

Evidence-backed. Government-proven. No direct competitor.

TECHNICAL FEASIBILITY

Tech Stack
High

React + FastAPI + PostgreSQL + Docker — all proven at India govt scale (CoWIN, GSTN, IRCTC)

ML Engine
High

Isolation Forest (unsupervised) — correct choice: no labeled fraud data exists in govt spending

Lapse Prediction
High

Weighted linear extrapolation — 100% auditable, officers verify every number, 3× Q4 weighting

Scale to 718 Dists
High

NIC operates 718+ district data centres; PFMS already processes 2M+ transactions/day

Regulatory Fit
Very High

Directly aligned with Digital India, PFMS expansion mandate, National AI Strategy 2021

MARKET & FINANCIAL VIABILITY

$21Bn

GovTech Market by 2027
12% CAGR | Gartner 2023

₹47.65L Cr

India Union Budget FY2024-25
Zero real-time watchdog today

966× ROI

ROI: ₹48,300 Cr recoverable vs ~₹50 Cr platform cost

0

Direct competitors with real-time AI budget monitoring

NITI Aayog 2022
+23% utilization

Digital expenditure tracking — highest-ROI govt intervention

Finance Ministry FY2024
61% by Q3

Capex only 61% utilized — 78% of road budget spent in last 6 weeks

18 months → 5 minutes | 100,000× faster anomaly detection | Break-even: 1–2 state contracts

BUSINESS MODEL

State-first beachhead. Multiple revenue streams. Path to profitability in 1–2 contracts.

State SaaS
₹5–15 L
per state / yr

28 states → ₹420 L ARR

District SaaS
₹25–50 K
per district / yr

718 districts → ₹359 L ARR

Ministry Licence
₹10–30 L
per ministry / yr

40+ ministries → ₹400 L+

Implementation
₹1–5 K
per deployment

One-time · high-margin

GO-TO-MARKET

01 State Beachhead
6–18 mo procurement vs 4 yrs central. Maharashtra, Rajasthan, Telangana as pilots.

02 Proof by ROI
One state = World Bank-citable case study → replicates to 28 states.

03 National Scale
28 states + 40 ministries + 718 districts via NIC infrastructure.

FINANCIAL CASE

Annual OpEx — **₹6.25–12.5 L**
Engineering + infra + compliance

Break-even — **1–2 contracts**
Single state deal covers full OpEx

Recoverable — **₹48,300 L/yr**
23% of ₹2.1L Cr lapse (NITI Aayog)

KEY PARTNERSHIPS

NIC
Govt infrastructure + credibility

NASSCOM GovTech
Access to 50+ state CIOs

World Bank / UNDP
Grant funding for low-cap states

Sources: NITI Aayog 2022; Gartner 2023; World Bank AP e-Procurement Case Study 2018; MeitY Digital India FY2024

PROJECTED IMPACT

Based on NITI Aayog research: digital expenditure tracking = +23% utilization improvement.

₹48,300 Cr

Recoverable annually (23% of ₹2.1L Cr lapse)

< 5 min

Anomaly detection vs 18-month audit lag

718

Indian districts platform can scale to

100%

Audit trail on every reallocation suggestion

COMPARABLE SUCCESS: Andhra Pradesh's e-Procurement system (2003) reduced processing time by 85%, saved ₹1,200 Cr in 3 years, and is now cited by World Bank as a model for GovTech ROI. Our platform applies the same principle — transparency + automation — specifically to budget execution monitoring.

RESEARCH & EVIDENCE

The problem is real, documented, and massive. Our platform addresses every finding below.

CAG India 2022
₹1.6 lakh crore in unspent scheme funds across Central Ministries. 42% of surveyed debts had zero spending in first two quarters.

₹1.6L Cr

Finance Ministry FY2024
Capital expenditure utilization was only 61% by Q3. Road transport ministry spent 78% of budget in last 6 weeks of fiscal year.

61% Q3 capex

World Bank (2021)
'Budget credibility' — ratio of actual to planned spending — averages 0.74 across developing nations. India scores 0.71 for capital expenditure.

0.71 credibility

NITI Aayog Report 2022
Districts with digital expenditure tracking show 23% better budget utilization vs districts without. Real-time monitoring is the single highest-ROI intervention.

+23% utilization

Transparency Intl. 2023
India ranks 93/180 on Corruption Perceptions Index. Procurement & public spending remain highest-risk areas. Automated monitoring is a proven deterrent.

93/180 CPI rank

THE ASK IS SIMPLE

₹2.1 lakh crore leaks out of India's budget every year.
We built the system that stops it.

🏗️ Built in 24 hours. Production-ready architecture. Real ML. Real DB. Real APIs.

📊 Backed by CAG, World Bank, NITI Aayog, Finance Ministry data.

🚀 One platform to serve 718 districts, 28 states, and every ministry in India.